

2.3 Differences to note with C#

1. The default access modifier of class members is private.
2. Instead of saying extends/implements, the class name and the list of classes/interfaces it extends/implements is separated with a colon (ala C++).
3. Java allows subclasses to override any of its visible methods. C# is more strict in the interests of encapsulation. Parent classes must mark overridable methods as virtual (ala C++).
4. C# requires that subclasses explicitly mark overridden methods with overrides.
5. Instead of super() as the first call in a constructor, C# uses C++-like initialiser syntax to invoke superclass constructors.
6. Instead of super to invoke methods of the base class, C# uses super to the same effect.
7. C# has a notion of a static class not present in Java that restricts said class

Lots of little differences to note but the fundamental idea is the same (and indeed the two sets of classes perform identically).

2.4 Hiding class members with new

As mentioned previously, we must mark methods in subclasses that override virtual methods in parent classes with overrides. If we wish to instead hide/replace a method (or any inherited members) in a parent class with a subclass version, we use new.

```
public class NewTest
{
    public class Parent
    {
        // Note: F is not virtual so it subclasses cannot
        // override it
        public void F() { Console.WriteLine("Parent"); }
    }
    class Child : Parent
    {
        // Hide Parent's F with our own F
        public new void F() { Console.WriteLine("Child"); }
    }
}

#endregion
```